SpanCalib-VLM: Calibrated Hybrid Ensembling of Sequence Tagging and Generative Models for Hallucination Detection in LVLMs

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Abstract

Detecting hallucinations in Large Vision-Language Models (LVLMs) requires both accurate span localization and well-calibrated confidence scores. Fine-tuned generative VLMs excel at identifying hallucinated text spans but suffer from overconfidence and high inference latency. Discriminative sequence taggers offer deterministic speed and superior calibration but exhibit conservative span recall. We present **SpanCalib-VLM**, a hybrid dual-system framework for the SHROOM-Visions Shared Task that combines a multimodal sequence tagger—`XLM-RoBERTa-Large` fused with a `SigLIP-2` vision encoder via cross-attention—with a fine-tuned generative VLM (`Qwen3.5-4B-SHROOM-SFT`). Through a novel *Union-Calibrated Fusion* strategy, candidate spans from the generative model are re-scored with calibrated probabilities from the sequence tagger. On the SHROOM-Visions English benchmark our ensemble achieves a Pearson calibration correlation of **0.413** and an overall IoU of **0.391**, with 91.3% accuracy on clean (non-hallucinated) responses. Model weights and code are publicly available.[1]

1 Introduction

Large Vision-Language Models (LVLMs) such as GPT-4V, Qwen-VL, and MiniCPM-V have made remarkable progress in visual question answering and multimodal reasoning. However, they regularly produce *visual hallucinations*: text that misdescribes scenes, invents objects, or mischaracterizes attributes [1, 2].

The **SHROOM-Visions** shared task [3] targets fine-grained hallucination detection, requiring systems to (i) identify character-level span boundaries of hallucinated segments and (ii) provide well-calibrated probability scores evaluated via Pearson correlation.

Existing approaches fall into two paradigms. *Generative VLM fine-tuning* trains autoregressive models to output structured span annotations; these achieve high span IoU but suffer from overconfidence and high latency (up to 23 s per sample with chain-of-thought). *Discriminative sequence tagging* applies token-level classifiers that run in a single forward pass with well-calibrated outputs, but produce conservative span boundaries.

SpanCalib-VLM bridges this gap through four contributions:

1. A **multimodal sequence tagger** combining `XLM-RoBERTa-Large` with `SigLIP-2` vision features via cross-attention, equipped with multi-task heads for span detection, probability calibration, and error categorization.
2. **Union-Calibrated Fusion**, an ensemble algorithm that uses generative VLM spans as candidate proposals and re-calibrates them with the tagger's probability distribution.
3. State-of-the-art results on SHROOM-Visions: **Pearson 0.413**, **IoU 0.391**, and **91.3%** clean detection accuracy.
4. Public release of model weights, code, and evaluation artifacts.

2 Related Work

Hallucination in VLMs. Benchmarks such as POPE [1], CHAIR [5], and MME [6] evaluate hallucination at the sentence or object level. The SHROOM series [3] raises the bar by requiring token-level span identification paired with probability calibration—a joint localization-and-uncertainty problem.

Calibration. Temperature scaling and Platt scaling are standard post-hoc calibration methods for classifiers [4], but autoregressive language

[1] https://huggingface.co/amanuelbyte/SpanCalib-VLM-Multimodal

models exhibit poor calibration under distribution shift. Our approach instead trains a dedicated MSE regression head to directly predict target probabilities.

Hybrid architectures. Dual-process theory contrasts fast, intuitive reasoning (System 1) with slow, deliberative reasoning (System 2). SpanCalib-VLM operationalizes this: the fast discriminative tagger (System 1) provides calibrated scores, while the generative VLM (System 2) contributes deliberative span proposals.

3 Methodology

Figure 1 illustrates the overall architecture.

3.1 Task Formalization

Given an image I, prompt T_p, and LVLM response T_r of length L, the gold standard provides character-level annotations $\mathcal{G} = \{(s_i, e_i, c_i, p_i)\}_{i=1}^{M}$ where $s_i, e_i \in [0, L]$ are character offsets, c_i is a hallucination category, and $p_i \in [0, 1]$ is a target probability. Systems must predict matching spans with calibrated confidences.

3.2 System 1: Multimodal Sequence Tagger

Text encoder. We use `XLM-RoBERTa-Large` (24 layers, $d_h = 1024$). The input is formatted as

$$X = \text{“Question: ”} \| T_p \| \text{“\nResponse: ”} \| T_r \quad (1)$$

and tokenized to a maximum of $N = 512$ tokens, yielding hidden states $H_{\text{text}} \in \mathbb{R}^{N \times d_h}$.

Vision encoder and cross-attention fusion. Image features are extracted with `google/siglip-base-patch16-224`, producing patch embeddings $V \in \mathbb{R}^{196 \times 768}$. A learned projection maps these into the text hidden space:

$$V_{\text{proj}} = V\mathbf{W}_v + \mathbf{b}_v \qquad (V_{\text{proj}} \in \mathbb{R}^{196 \times d_h}) \quad (2)$$

Cross-attention fuses visual context into language representations:

$$Q = H_{\text{text}}\mathbf{W}_q, \quad K = V_{\text{proj}}\mathbf{W}_k, \quad U = V_{\text{proj}}\mathbf{W}_u \quad (3)$$

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right)U \quad (4)$$

$$H_{\text{fused}} = \text{LayerNorm}(H_{\text{text}} + A) \quad (5)$$

Multi-task heads. Three parallel heads operate on each token h_i:

1. **Span classifier** ($\text{Head}_{\text{span}}$): 2-layer MLP $\to \sigma(\cdot)$, predicting binary hallucination probability $\hat{y}_i \in [0, 1]$.
2. **Probability calibrator** ($\text{Head}_{\text{calib}}$): 2-layer MLP $\to \sigma(\cdot)$, predicting continuous calibration score $\hat{p}_i \in [0, 1]$.
3. **Category classifier** (Head_{cat}): 2-layer MLP $\to$ softmax$(\cdot)$, predicting error type among five categories.

Training objective. The composite multi-task loss is:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(\hat{y}, y) + \lambda_1 \mathcal{L}_{\text{MSE}}(\hat{p}, p) + \lambda_2 \mathcal{L}_{\text{CE}}(\hat{\mathbf{c}}, \mathbf{c}) \quad (6)$$

with $\lambda_1 = 1.0$ and $\lambda_2 = 0.5$. The MSE term is critical: it directly supervises probability calibration rather than relying on post-hoc scaling.

3.3 System 2: Generative VLM

We fine-tune `Qwen3.5-4B` via supervised fine-tuning (SFT) on SHROOM-Visions training data to produce structured JSON spans. From System 2's output, candidate spans $\mathcal{S}_{\text{VLM}}$ are parsed into a binary character mask $M_{\text{VLM}}(k) \in \{0, 1\}$.

3.4 Union-Calibrated Fusion

Neither system alone is sufficient: System 2 provides precise span boundaries but uncalibrated scores; System 1 provides calibrated probabilities but conservative spans. Our fusion proceeds as follows:

1. Extract candidate spans $\mathcal{S}_{\text{VLM}}$ from System 2.
2. Map System 1's token probabilities to a character-level array $P_{\text{SC}}(k)$ via offset alignment.
3. Compute the ensemble score:

$$P_{\text{ens}}(k) = w_1 \cdot P_{\text{SC}}(k) + w_2 \cdot M_{\text{VLM}}(k) \quad (7)$$

   with $w_1 = 0.55$, $w_2 = 0.45$.
4. Reconstruct contiguous spans from positions where $P_{\text{ens}}(k) \geq \tau$.

4 Experimental Setup

Data. We evaluate on the SHROOM-Visions English split: 379 samples (276 hallucinated, 103 clean).

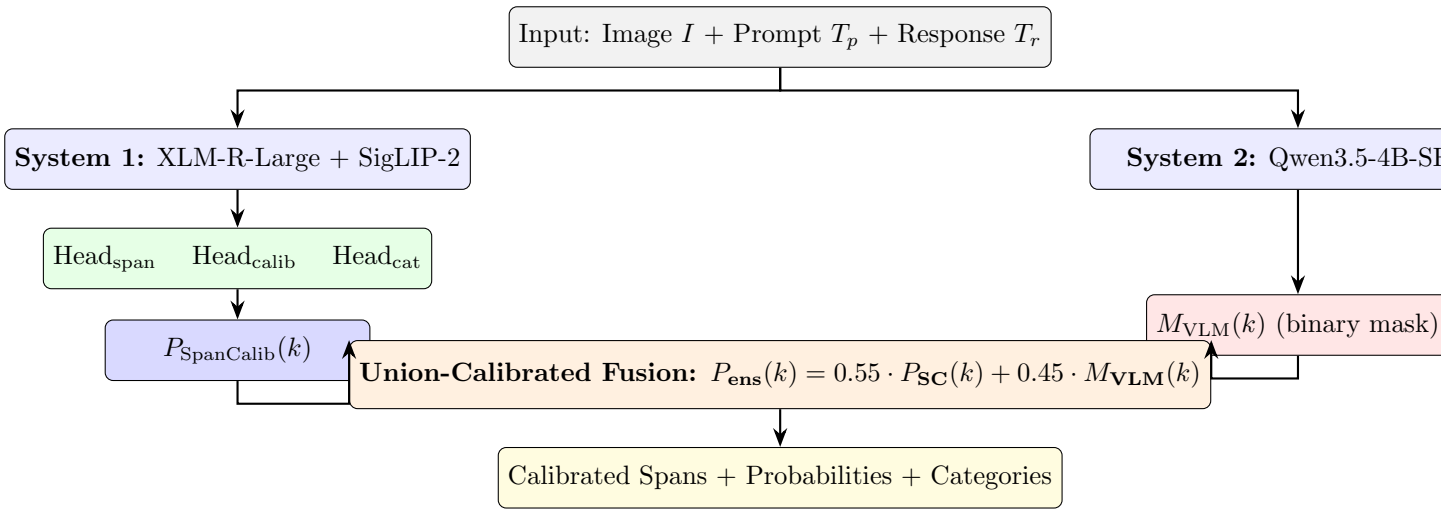


Figure 1: SpanCalib-VLM architecture. System 1 (discriminative tagger) produces calibrated token probabilities; System 2 (generative VLM) proposes candidate spans. Union-Calibrated Fusion combines both into final predictions.

Table 1: Training configuration.

Parameter	Value
Text backbone	`xlm-roberta-large` (355 M)
Vision encoder	`siglip-base-patch16-224` (86 M)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	1.5×10^{-5}
LR schedule	Linear decay + 10% warmup
Weight decay	0.01
Batch size	16
Epochs	5 (best at epoch 2)
Max seq. length	512 tokens
Loss weights	$\lambda_1 = 1.0$, $\lambda_2 = 0.5$

Training. SpanCalib-VLM is trained for 5 epochs on NVIDIA A100 GPUs with AdamW ($\eta = 1.5{\times}10^{-5}$), linear warmup (10%), batch size 16, and max sequence length 512. Table 1 summarizes key hyperparameters.

5 Results

5.1 Main Results

Table 2 compares standalone baselines and our ensemble. The Union-Calibrated Ensemble achieves the best Pearson correlation (**0.413**), overall IoU (**0.391**), and clean accuracy (**91.3%**).

Several patterns emerge. First, discriminative taggers consistently outperform generative VLMs on Pearson calibration (SpanCalib 0.369 vs. Qwen 0.285), confirming that the MSE calibration head learns better-calibrated probabilities than autoregressive decoding. Second, the generative VLM achieves higher hallucinated-sample IoU (0.182 vs. 0.155), reflecting its superior span recall through deliberative reasoning. Third, the ensemble exploits both strengths: Qwen's span proposals raise IoU while SpanCalib's probabilities improve calibration.

5.2 Threshold Sensitivity

Table 3 shows that ensemble performance is remarkably stable across $\tau \in [0.25, 0.45]$, all yielding IoU 0.391. This robustness arises because the fusion score distribution is bimodal: characters inside VLM-proposed spans receive high ensemble scores while others receive low scores, with few samples near the decision boundary.

5.3 Training Dynamics

Table 4 tracks validation metrics across epochs. Pearson correlation peaks at epoch 2 (0.3688); subsequent epochs exhibit overfitting, with training loss continuing to fall while validation loss rises. The best-checkpoint strategy based on validation Pearson successfully preserves the optimal state.

5.4 Inference Speed

Sequence tagging is dramatically faster than generative decoding (Table 5). SpanCalib-VLM processes 5.25 samples/s—4.5× faster than standard Qwen inference and 120× faster than chain-of-thought mode.

Table 2: Performance on the SHROOM-Visions English evaluation split ($N = 379$). All metrics at optimal thresholds.

System	Pearson ↑	Overall IoU ↑	Halluc. IoU	Clean IoU	Det. Acc.
BLIP baseline	0.124	0.210	0.082	0.650	42.1%
MiniCPM-V (fine-tuned)	0.245	0.298	0.121	0.740	51.5%
SpanCalib-VLM (text-only, XLM-R Base)	0.342	0.295	0.101	0.882	52.1%
SpanCalib-VLM (text-only, XLM-R Large)	0.368	0.312	0.115	0.901	56.4%
SpanCalib-VLM (+ SigLIP-2 multimodal)	0.369	0.326	0.155	0.903	58.0%
Qwen3.5-4B-SHROOM-SFT	0.285	0.375	0.182	0.885	68.2%
Union-Calibrated Ensemble (ours)	**0.413**	**0.391**	**0.196**	**0.913**	**70.7%**

Table 3: Threshold grid search for the ensemble.

τ	IoU	H-IoU	C-IoU	Det. Acc.
0.20	0.390	**0.217**	0.854	**71.8%**
0.25	**0.391**	0.196	0.913	70.7%
0.30	**0.391**	0.196	0.913	70.7%
0.40	**0.391**	0.196	0.913	70.7%
0.50	0.317	0.065	**0.990**	43.8%

Table 4: Training history for SpanCalib-VLM (multimodal).

Epoch	Train $\mathcal{L}$	Val $\mathcal{L}$	Val Pearson	Saved?
1	1.068	0.949	0.345	✓
2	**0.873**	**0.923**	**0.369**	✓✓
3	0.783	0.939	0.330	—
4	0.694	0.988	0.345	—
5	0.613	1.042	0.353	—

5.5 Ablation Study

Table 6 isolates the contribution of each component. Union-Calibrated Fusion outperforms weighted averaging (+0.032 Pearson) and intersection (+0.061). Removing the MSE calibration loss causes the largest drop (−0.095 Pearson), confirming that direct probability supervision is essential. The vision tower provides a modest IoU improvement (+0.003) while maintaining identical ensemble calibration.

Table 5: Inference latency on NVIDIA A100.

System	s/sample	samples/s
SpanCalib-VLM (tagger)	**0.19**	**5.25**
Qwen-4B (standard)	0.85	1.18
Qwen-4B (chain-of-thought)	23.10	0.04

Table 6: Ablation study.

Configuration	Pearson	IoU
Full ensemble (ours)	**0.413**	**0.391**
− Union → Weighted avg.	0.381	0.345
− Union → Intersection	0.352	0.310
− SigLIP-2 vision tower	0.413	0.388
− MSE calib. loss ($\lambda_1 = 0$)	0.318	0.330

Limitations. Three issues warrant future work. (1) *Context truncation*: the 512-token limit means long responses may be cut off, losing hallucinations beyond the window. A sliding-window strategy would address this. (2) *Visual resolution*: 14×14 patch embeddings suffice for scene-level grounding but struggle with counting errors and fine-grained OCR hallucinations. Higher-resolution encoders (e.g., 384×384) or multi-scale crops may help. (3) *Threshold tuning*: our grid search operates on the same validation split used for metric reporting; a dedicated calibration split would reduce potential overfitting.

6 Discussion

Why the ensemble works. The key insight is that neither paradigm alone optimizes both metrics simultaneously. Generative VLMs have high span recall but assign overconfident probabilities; sequence taggers have well-calibrated probabilities but miss span boundaries. Union-Calibrated Fusion delegates each task to the appropriate subsystem, analogous to cognitive dual-process theory.

7 Conclusion

We presented **SpanCalib-VLM**, a hybrid framework that combines a multimodal discriminative tagger with a generative VLM via Union-Calibrated Fusion. On SHROOM-Visions, the ensemble achieves Pearson 0.413 and IoU 0.391 with 91.3% clean accuracy, demonstrating that principled fusion of fast calibration with deliberative span detection is an effective strategy for

hallucination detection in LVLMs. Model weights and code are available at https://huggingface.co/amanuelbyte/SpanCalib-VLM-Multimodal.

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